Parth — Software Team Spec

Parth is the flagship adult companion (ages 20–45): **AI chief-of-staff, mentor, expert council**, with a 3.5" touch glance face. This document lets the **software team build the AI brain, integrations and app in parallel with the hardware**, by reusing the Xelulu gateway contract and layering Parth-specific features on top.

The hardware spec is deliberately thin here — it is covered by `Parth_BOM_ESP32S3_v4.xlsx` + the firmware. **This document is about AI features and what the software team must deliver.**

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1. How Parth differs from Xelulu (one page)

Thing	Xelulu	Parth
Audience	Kids 1–12	Adults 20–45 (flagship)
Form	Mobile egg with wheels + arms + visor	Static desk orb with touch face
Face	2" pixel-LED visor, no touch	**3.5" resistive-touch HMI** (UART smart-screen)
Primary input	Voice + touch zones + motion	Voice + touch screen + optional TTP223 taps
On-device perception	Wake word, cry detection	Wake word only
Movement	Yes (2 wheels, ToF)	**None** (static)
Camera	None	**None** (ESP32-S3 build; SBC build later for webcam)
Phone calls	—	**Yes**, via external BT-Classic module (HFP)
Smart home	—	**Matter over Wi-Fi**
Battery	2S	1S2P

Reuses from Xelulu, zero changes: the gateway contract (§3 of the Xelulu spec), auth model, cloud infrastructure, voice pipeline (STT→LLM→TTS), app accounts/households, OTA pipeline.

New in Parth (software team work): everything in §4 — the AI feature catalogue (chief-of-staff, meetings, research, drafting, mentor, council, smart home), the HMI display GUI, the BT-call bridge, Matter integration, and the connections/permissions model.

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2. Scope & parallel delivery

The same rule as Xelulu: freeze the device gateway API first, build to it. Parth adds a small delta on top of the shared gateway (§3 of this doc), not a new protocol. Everything above the gateway is identical in shape —

new feature services, new UI, new integrations — but the transport, auth, device identity and message envelopes are the same.

Software team owns (this doc):

- **Chief-of-staff services** — calendar / mail / tasks integrations, drafting, meeting prep, follow-up tracking, personal admin.
- **Meeting companion** — recording consent, transcription, speaker separation, summary + action-item extraction.
- **Research agent** — multi-step web research with source attribution.
- **Expert Council + Mentor + Rehearsal coach** — multi-persona LLM orchestration, goals / weekly reviews / decision journal.
- **Smart home** — Matter-over-Wi-Fi controller service.
- **Phone-call bridge control** — commanding the external BT module via firmware.
- **HMI GUI pages** — designed in the SmartElex HMI editor, stored on the display's own micro-SD; the firmware only sends page IDs + variable values over UART.
- **Parth module in the shared app** — connections, rules, approvals, council memos, meeting transcripts.
- **Device simulator (Parth profile)** — a feature flag on the Xelulu simulator (§9 of that spec) that emits Parth events (touch screen, BT call state) and accepts Parth commands.

Hardware team owns: the ESP32-S3 gateway firmware, HMI display wiring, BT module wiring + AT-command firmware, Matter radio stack on ESP32-S3, I2S audio pipeline, power, enclosure.

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3. Gateway deltas vs. Xelulu

Reuse Xelulu §3.1 (transport, auth, versioning, heartbeat, QoS) and §3.2 (topic shape; `xelulu/` becomes `parth/{device_id}/...`). Only the following message types change.

3.1 New / changed events (device → cloud)

Event `type`	`data`	Replaces / adds
`screen_touch`	`{ page: "home", target: "card_next_meeting", gesture: "tap"	"long"
`tap_zone`	`{ zone: "head_top"	"head_left"
`bt_call`	`{ state: "incoming"	"active"
`focus`	`{ action: "start"	"end"
`wake`	same shape as Xelulu	Reused
`motion`	**drop** `pickup/shake/upside_down` (static device); keep nothing	Removed
`touch`	**drop** (zones replaced by `tap_zone` / `screen_touch`)	Removed

3.2 New / changed commands (cloud → device)

Command `action`	`data`	Purpose
`screen`	`{ page: "home"	"meeting_card"

Command `action`	`data`	Purpose
`screen_toast`	`{ text: "Meeting in 10", kind: "info"	"warn"
`bt_call`	`{ cmd: "answer"	"hangup"
`focus`	`{ cmd: "start"	"end", minutes: 25, rules: {...} }`
`matter`	`{ cmd: "scene"	"on"
`say` / `animate` / `play` / `set_mode` / `config` / `ota`	same as Xelulu	Reused (animate is lightweight — face expression swap on the HMI)
`gesture` / `move`	**drop** (no arms, no wheels)	Removed

3.3 Status

Same as Xelulu, plus two fields:

```
{ "..." : "...", "bt_module": "ready"|"not_paired"|"in_call", "matter_hub": "ready"|"commissioning"|"error" }
```

3.4 Freeze process

Same as Xelulu — both leads sign off this §3 before anyone writes feature code. Changes bump `proto_ver`.

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4. AI feature catalogue — the software work

Every row maps a catalogue feature (PA-* IDs from the Product Feature Catalogue §6) to what the software team must build. Priorities come from the catalogue; **P0 = must have at launch**.

4.1 Chief of staff (catalogue §6.1)

ID	Feature	What software must build	Priority
PA-C01	**Glance screen**	HMI pages: status strip (time/weather/temp always on), rotating main cards (next meeting w/ countdown, agenda, top tasks, commute time, focus timer). Cloud pushes `screen` cmds with vars every 15 s or on event.	P0
PA-C02	**Morning / evening brief**	Scheduled briefing service: pulls user data (calendar, mail priority, weather, news if opted in, tasks) → LLM composes brief → TTS plays voice + `screen` cmd shows cards. User sets time + content categories.	P0
PA-C03	**Calendar command**	Integrations: Google Calendar + Microsoft 365 (OAuth). LLM function-calling tools: `list_events`, `create_event`, `move_event`,	P0

ID	Feature	What software must build	Priority
		`cancel_event`, `find_free_slot`, `propose_times`. **Mandatory confirmation step** before any action that touches other people.	
PA-C04	**Mail triage**	Integrations: Gmail + Outlook. Services: priority classifier (user rules + VIP list), summarizer, draft composer. `send` requires explicit "Yes send it" voice or screen confirmation.	P0
PA-C05	**Tasks + follow-ups**	Integrations: Todoist, Notion, Google Tasks, Microsoft To-Do (configurable). Follow-up tracker: watches sent mail → cron check "if no reply by Thursday" → remind on glance screen.	P0
PA-C06	**Meeting prep packs**	15 min before each meeting: pull attendees → cross-reference last-interaction mails + notes → fetch linked docs → LLM composes a 1-pane prep card → pushed to glance screen + app.	P0
PA-C07	**Research agent**	Multi-step web research tool: LLM agent with `search`, `open_url`, `extract`, `cite`. Produces a markdown report with source links to the app or user's mail. Rate-limit, cost-cap per request.	P0
PA-C08	**Drafting (user's style)**	Style-sample ingestion flow (user approves which samples go in). Prompt-engineered or fine-tuned writer per user. Outputs to app or connected doc tools (Google Docs, Notion).	P0
PA-C09	**Personal admin**	Mail parsers for bills, renewals, travel itineraries, package tracking; voice-captured expense log; reminders surfaced on glance screen.	P1
PA-C10	**Phone bridge (notifications)**	Companion phone app (Android first): read notifications w/ user permission → forward summary to cloud → voice readout; voice-dictated reply sent back via phone. iOS limited to calendar/mail/reminders.	P1
PA-C11	**BT speakerphone**	Pair user's phone to the BT module (one-time via app). Call audio routes through the module's own mic + Parth's speaker. Firmware controls answer/hangup via AT commands over UART; cloud shows the call-screen page.	P1

4.2 Meeting companion (§6.2)

ID	Feature	What software must build	Priority
PA-M01	**In-room notes**	Voice intent → cloud opens a long-form audio session; **cloud emits a `screen` cmd setting the face to "Recording" with a red ring + spoken announcement**; stream mic audio up; store as transcription-only (audio deleted after processing by default).	P0
PA-M02	**Speaker tracking**	Mic-array direction estimation only (no camera on this build). Face page animates a "looking toward" indicator; actual head turn is a Parth-SBC-later feature. Diarization in the transcription service.	P0*

ID	Feature	What software must build	Priority
PA-M03	**Summary + action items**	Within 2 min of meeting end: LLM extracts decisions, action items with owners + due dates, open questions. User approves before any resulting mail/task/calendar action is executed.	P0
PA-M04	**Live assist**	While recording, user whispers "Parth, what did we agree last time?" → quiet retrieval over past meeting index → shows card on the face (not spoken aloud mid-meeting). Also time-keeping toasts ("10 min left").	P1
PA-M05	**Consent controls**	Any room participant can say "Parth, stop recording" — wake-word recogniser must handle this phrase from any voice, not just the owner. Retention default: delete raw audio after transcription; user-configurable.	P0

* `PA-M02` is reduced to mic-array direction only on the ESP32-S3 build (no camera). Full speaker tracking returns in an SBC Parth variant.

4.3 Webcam & video calls (§6.3) — DEFERRED

ESP32-S3 Parth has **no camera**. The following are **dropped from P0 and move to a future SBC Parth build**:

- PA-W01 USB webcam
- PA-W02 USB speakerphone (standalone — Parth works as a Bluetooth speakerphone instead, see PA-C11)
- PA-W03 Auto-framing / speaker tracking by camera
- PA-W04 Camera privacy indicators (nothing to protect)

Kept, voice-only variant: PA-W05 Family calls (voice-only to another Parth / Sarthi / Xelulu via the household intercom in PA-H04).

Call this out in the app: the ESP32-S3 Parth is marked "voice edition"; webcam features are shown as "coming to Parth Pro".

4.4 Focus & wellbeing (§6.4)

ID	Feature	What software must build	Priority
PA-F01	**Bird's Eye focus**	Focus session state machine (start/pause/end) → `focus` cmd + `screen` to focus-page + optional phone DND via companion app. VIP/urgent-rule bypass evaluates each incoming notification before it is relayed.	P0
PA-F02	**Walk-up handling**	**Dropped** on ESP32-S3 Parth — no ToF, no camera. Replace with "tap head to leave a note" via TTP223 or the touch screen.	P1 → simplified
PA-F03	**Healthy work habits**	Timers for hydration, 20-20-20 eye rest, stand-and-stretch. Posture nudge (camera-based) **dropped** here; keep hydration/eye/stretch.	P1 (reduced)

ID	Feature	What software must build	Priority
PA-F04	**Wind-down**	End-of-day ritual: LLM asks pre-set reflection prompts → captures open loops → sets tomorrow's first task → good-night face.	P1

4.5 Mentor & expert council (§6.5)

ID	Feature	What software must build	Priority
PA-E01	**Expert Council**	Multi-agent LLM orchestration: up to 6 named lenses (CFO, lawyer, engineer, marketer, investor, customer, or user-defined). Each gets the user's plan + its own system prompt; outputs independent views; Parth synthesises a memo. **Memo delivered to the app, not read aloud in full.** Must display "AI perspectives, not professional advice."	P0
PA-E02	**Mentor mode**	Goals service (quarterly + weekly); weekly review flow; decision journal (user dictates decisions → stored with context → surfaced later for reflection); kind commitment tracker (never shaming).	P0
PA-E03	**Rehearsal coach**	Role-play LLM mode (interviewer / investor / counterpart); feedback on structure, pace (words/min from mic), filler words. **Eye-contact feedback dropped** (no camera).	P0 (reduced)
PA-E04	**Skill plans**	LLM-generated learning plans + resource curation + spaced-practice scheduler.	P1
PA-E05	**Student chief of staff**	Semester planner, assignment deadline tracker, lecture/paper summariser (user pastes/uploads), thesis coach prompts, placement-prep practice sets, internship tracker. Shares infra with PA-C* and PA-E02.	P0

4.6 Home & life (§6.6)

ID	Feature	What software must build	Priority
PA-H01	**Smart home (Matter)**	Cloud Matter controller service **or** on-device Matter commissioner (ESP32-S3 supports Matter over Wi-Fi — Thread deferred, no Thread radio in BOM). Scenes, routines, status cards on the glance screen.	P0 (Wi-Fi only)
PA-H02	**Music & media**	Integrations: Spotify, YouTube Music, podcasts. Audio streamed via cloud to the device's I2S speaker (or BT module for higher quality music).	P1
PA-H03	**Timers, alarms, reminders, lists**	Core assistant primitives; must work offline (firmware timers + cached alarms).	P0
PA-H04	**Household intercom**	Voice-only call to another household Companion device (Xelulu / Sathi / Sarthi / another Parth). Reuses the cloud gateway — one device publishes an audio turn, the other subscribes. Permission-gated.	P1

ID	Feature	What software must build	Priority
PA-H05	**Everyday knowledge**	General LLM Q&A, translation (10 Indian + major world languages per catalogue §3), cooking help, unit/currency conversions, news explained.	P0

4.7 Offline mode (§6.7)

Must work with no internet (firmware + cached content):

- Wake word, touch screen, TTP223 taps.
- Timers, alarms, reminders (previously-synced only; new ones queued).
- Glance face shows a cached "last known" morning brief + offline indicator.
- BT calls (the module itself works offline over the paired phone).
- Smart home: Matter commands can work locally if the Matter stack is on the device (no cloud needed for local scenes).
- Offline line: "*I can't reach the brain right now — your timers, calls and lights still work.*"

Re-sync on reconnect: flush queued events (focus sessions, touch events, voice captures that fell through), fetch missed briefs, apply pending config / OTA.

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5. HMI display GUI contract

The SmartElex 3.5" HMI is a **smart display** — it renders its own GUI pages from assets on its own micro-SD card, driven by UART commands from the ESP32-S3. **Software team designs these pages in the SmartElex HMI editor and ships the resulting SD-card image as part of the OTA bundle.**

5.1 Required pages (P0)

Page name	Elements	Pushed vars from cloud
`home`	Status strip (time, weather, temp), rotating main card	`card_type`, `card_title`, `card_body`, `countdown_sec`
`brief`	Morning / evening brief as swipeable cards	`cards[]` (title, body)
`meeting_card`	Next meeting focus card	`title`, `time_in`, `attendees`, `location`
`focus`	Full-screen focus timer, minimal	`minutes_remaining`, `label`
`call`	Incoming / active call UI	`caller`, `state`, `duration`
`council_memo`	Scrollable AI memo view	`memo_md`
`recording`	Red ring + "Recording" + stop button	`elapsed`, `can_stop`
`offline`	Offline notice + what still works	—
`pairing`	QR code + pairing status	`code`, `step`

5.2 Interaction

- Each tappable region on each page has a **stable target name**; firmware reports it as `screen_touch` → cloud (§3.1).
- No pixel coordinates cross the gateway — only logical targets.
- Face expressions (`animate`) are a small enum of face frames stored on the SD-card and switched by UART.

5.3 OTA

GUI assets ship as a single versioned bundle (`hmi_assets_v{n}.bin`). OTA to the device includes both firmware and (optionally) a new HMI bundle; the firmware writes the bundle to the HMI's SD-card over its UART file-transfer protocol.

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6. Connections, OAuth & the permissions model

Parth is only useful once it has access to the user's mail, calendar, tasks, docs, smart home, and (optionally) phone. The software team must build a **first-class connections flow** in the app.

6.1 Supported connections (P0)

Connection	Purpose	Scopes requested
Google (Calendar, Gmail, Tasks, Drive)	Chief of staff, drafts	per-service, incremental
Microsoft 365 (Outlook, Calendar, To-Do, OneDrive)	Chief of staff, drafts	per-service, incremental
Notion / Todoist	Tasks + notes	read + write (optional)
Spotify / YouTube Music	Music	playback
Matter smart-home	Lights, scenes	commissioner
Phone companion app	Notifications bridge	OS-level

6.2 Permissions model

Per-connection, the user grants one of three levels:

- **Read** — Parth may see this data (brief, prep packs, triage).
- **Draft** — Parth may compose, but must ask before sending.
- **Act** — Parth may act without asking (irreversible changes still always ask).

Default is **Draft**. Nothing irreversible (send, delete, pay, buy) ever skips confirmation, regardless of level.

6.3 Rules engine

User-editable rules shown in plain language in the app:

- **VIPs** — people whose mail/calls always break focus.
- **Urgent rules** — keywords, labels, sources that bypass DND.
- **Focus rules** — what Parth may interrupt me for.
- **Working hours** — when `act` level is allowed.

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7. Data models (deltas from Xelulu §6)

```
UserProfile      { user_id, display_name, voice_sample_id, writing_style_id,
                   working_hours{}, vip_list[], urgent_rules[], tz }
Connection       { connection_id, user_id, provider, scopes[], permission_level,
                   oauth_token (encrypted), status, last_sync }
Meeting          { meeting_id, user_id, calendar_event_id, started_at, ended_at,
                   recording_consent, transcript_id, summary, action_items[] }
Transcript       { transcript_id, meeting_id, segments[], speakers[],
                   retention: {audio: "deleted"|"kept_30d", text: "kept"} }
ActionItem       { item_id, meeting_id, owner, text, due, status, applied_to:
                   calendar|mail|tasks|none }
Goal             { goal_id, user_id, period: "q"|"w", title, checkins[] }
DecisionEntry    { entry_id, user_id, decided_at, context, choice, revisit_at? }
CouncilMemo      { memo_id, user_id, question, lenses[], views[], synthesis }
MatterDevice     { device_id, user_id, matter_node, type, room, capabilities[] }
Rule             { rule_id, user_id, kind: vip|urgent|focus|hours, body }
```

`Device` + `Household` + `ChildProfile` (treat Parth owner as a `User` not a `ChildProfile`), `Session`, `Event`, `Consent` — reused from Xelulu §6.

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8. Voice & conversation pipeline

Same as Xelulu §7, with Parth-specific additions:

1. **Wake word** on-device (same as Xelulu).
2. **Barge-in** — user can interrupt Parth mid-sentence; audio WS supports duplex.
3. **Tool-use loop** — the LLM has function-calling tools for every integration (`calendar.create_event`, `mail.send_draft`, `research.search`, `matter.scene`, `focus.start`, etc.). Each tool invocation that *writes* goes through the confirmation layer (§6.2).
4. **Multi-turn planning** — for research / drafting / council, the agent may take minutes. During long runs the glance screen shows "working on X" with progress.
5. **Latency budget** — same ≤1.2 s wake→first audio.

Multiple personas (Expert Council, Rehearsal Coach) are the same STT→LLM→TTS pipeline but with different system prompts / TTS voices. No new transport.

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9. Device simulator (Parth profile)

Extend the Xelulu simulator (§9 of that spec) with a Parth profile toggle:

- Emits `screen_touch`, `tap_zone`, `bt_call`, `focus` events via a web UI.
- Accepts `screen`, `screen_toast`, `bt_call`, `focus`, `matter` commands and renders a mock HMI (a small HTML pane showing the current page + vars).
- Simulates BT-call state transitions (ringing → active → ended).
- Simulates Matter device acks (lights on/off, scenes).
- Offline toggle for queue-and-resync tests.

Deliverable in the same repo; CI uses it for feature end-to-end tests.

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10. Safety, privacy & compliance (adult)

Different from Xelulu — this is an adult product with sensitive data (mail, calendar, meetings, finances).

- **Explicit consent for every connection.** OAuth scopes shown in plain language; revocable per-connection anytime.
- **Confirmation before irreversible actions.** Send, delete, pay, buy, book, share — always ask aloud + on the screen; voice-only confirmation requires a verified wake-phrase ("Parth, confirm").
- **Meeting recording consent.** Recording never starts silently (PA-M05). Any voice in the room can say "Parth, stop recording." Default retention: audio deleted after transcription.
- **Memory is user-visible.** Everything Parth "remembers" (preferences, decisions, approved style samples) is listed in the app and deletable per-item.
- **Household-locked voice samples** (writing style, voice-clone if used).
- **No open internet to the user** that doesn't go through the research agent's citation layer — Parth answers with sources, not opinions presented as facts.
- **Crisis protocol** (PA-E, personal topics) — calm response, encourages talking to a trusted person, shares helplines (India: Tele-MANAS 14416). No alerts to anyone else without explicit user choice.
- **Expert Council disclaimer** — every memo is clearly labelled "AI perspectives, not professional advice."
- **Encryption** — OAuth tokens, transcripts, memory encrypted at rest; access scoped by user.

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11. App module — Parth

Adds to the shared household app:

Area	Functions
Today	Agenda, tasks, follow-ups, meeting summaries, focus stats
Meetings	Transcripts, summaries, decisions, action items, approve-and-send
Council & mentor	Council memos, goals, weekly reviews, decision journal
Research & drafts	Reports, drafts, sources, export to connected tools
Connections	Per-service OAuth, permission levels, scopes, revoke
Rules	VIPs, urgent rules, focus rules, working hours, act-without-asking defaults (default: nothing irreversible)
Privacy	Memory list + delete, retention settings, recording policy, data export, account deletion
Device	Mic status, Matter hub status, BT pairing, updates, diagnostics

Role: single-user owner (unlike Xelulu's parent/caregiver/grandma split). Household members may each own their own Parth; devices intercom via PA-H04.

12. Milestones & parallel delivery plan

Software workstreams that proceed with **zero hardware**, then integration gates.

Phase	Software (no hardware needed)	Gate
M0 — Contract delta freeze	Sign off §3 of this doc (events + commands + HMI pages list)	§3 frozen, `proto_ver` 2
M1 — Simulator + gateway (Parth profile)	Extend Xelulu simulator with Parth events/commands; mock HMI pane	Sim connects, screen/screen_touch round-trip
M2 — Voice pipeline + tool-use loop	Wake→STT→LLM(+tools)→TTS; confirmation layer; barge-in	Live tool-calling conversation via sim
M3 — Chief of staff core	Google + Microsoft connections, PA-C01/02/03/04/05/06 end-to-end against test accounts	User can get a brief + triage mail + create a calendar event via sim
M4 — Meeting companion	PA-M01/03/05: recording session, transcription, summary + action-item extraction, consent flow	Mock meeting → summary + action items reviewed in app
M5 — Mentor + Council + Research + Drafting	PA-C07/08, PA-E01/02/03/05	Council memo generated end-to-end; research report with sources
M6 — HMI GUI bundle v1	Pages 5.1 designed in HMI editor; OTA bundle format; the simulator renders the pages	Any page + vars reproduce on the mock HMI faithfully
M7 — Smart home + BT calls	PA-H01 Matter-over-Wi-Fi; PA-C11 BT bridge via AT-command firmware stub	Toggle a Matter light; answer a mock BT call via screen
M8 — Parent app Parth module	§11 screens	User can drive the simulator end-to-end from the app
M9 — Safety & privacy audit	§10 full suite — connection revoke, retention, meeting consent, confirmations	Safety test suite green
M10 — Integration	Swap simulator for real device (hardware reaches its own M-gate)	Real device: brief + meeting + Matter scene + BT call
M11 — Pilot	OTA pipeline, telemetry, field fixes	Stable multi-day runtime

Integration (M10) is low-risk **because** everything above the gateway was built and tested against the simulator — the only new variable is the physical device honoring the extended contract.

13. What is explicitly NOT in this version (software team: do not build)

- Any camera-dependent feature (PA-W01–W04, posture nudge, eye-contact feedback, walk-up detection by camera, speaker tracking by camera). Deferred to a future SBC Parth variant.
- Thread radio (Matter-over-Thread). Only Matter-over-Wi-Fi.
- Local LLM / on-device NLP beyond wake word. Everything else is cloud.
- Multi-user profiles on a single Parth. One owner per device (household intercom still works across devices).

This spec covers Parth (ESP32-S3 cloud-gateway build, ages 20–45). It layers on the Xelulu spec — the gateway, auth, cloud infra, voice pipeline, OTA and app skeleton are shared. Sathi/WithU and Sarthi will follow the same pattern (reuse + delta spec).